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DS4002

4/30/2026

CS3: Coffee Shops, Business Licenses, and Rising Rent

Can public data reveal early signs of neighborhood change?

You are working as a junior data analyst for a community development organization in Chicago. Residents have raised concerns that new businesses may be connected to rising rents in their neighborhoods. Some community members point to new restaurants, nightlife spots, retail stores, and lifestyle businesses as signs that neighborhood change is accelerating. Your team has been asked to investigate whether public data supports this concern.

Your task is to study the relationship between business openings and rent growth across Chicago ZIP codes. Instead of relying only on anecdotes, you will use two real-world datasets: Chicago business license records and Zillow rent price data. The project combines these sources by ZIP code and month to create a time-based dataset that tracks rent prices, business openings, business categories, and monthly rent growth.

This case study places you in the role of an urban data analyst. You will clean and interpret messy public records, generate exploratory visualizations, and explain what the data can and cannot tell us about neighborhood change. The project's cleaning script standardizes ZIP codes, converts Zillow data into a monthly format, filters for Chicago-area ZIP codes, counts business openings by month, and calculates rent growth.

Your final deliverable will be a short analytical memo or slide deck for a non-technical audience. Your goal is to answer the question:

Do business openings appear to be associated with rent growth in Chicago ZIP codes?

You are not expected to prove that new businesses cause rent increases. Instead, your job is to explain whether there is an observable relationship, what patterns appear in the visualizations, and what limitations should be considered before making policy claims.

By the end of this case study, you should understand how data cleaning, geographic units, time periods, and visualization choices shape the conclusions we draw from urban data.